Multimodal Data Fusion in IoMT For Early Disease Detection using Deep Learning with Generated Data

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***Abstract*—In healthcare, Electronic Health Records (EHRs) and continuous patient monitoring data are restricted by privacy regulations like GDPR in France and HIPAA in the United States. To apply Deep Learning in Healthcare, we developed a rule-based synthetic data generation program that simulates a real-world Internet of Medical Things (IoMT) environment. The pipeline generated 200,000 synthetic patient records, including continuous vital signs, demographic history, and laboratory biomarkers across 24 distinct medical pathologies. We benchmarked two Deep Learning architectures: an Early Fusion Multi-Layer Perceptron (MLP) and a Contextual Fusion Tabular Transformer. Our Experimental results prove that the synthetic data poses a challenging classification task, with models achieving high accuracy, F1-score, and ROC-AUC through effective multimodal feature fusion. We also integrated SHapley Additive exPlanations (SHAP), which confirmed that the neural networks learned genuine physiological rules programmed into the generation engine, validating our synthetic dataset as competent for real-world IoMT predictive modeling.**

***Index Terms*—Synthetic Data Generation, Electronic Health Records (EHR), Internet of Medical Things (IoMT), Multimodal Data Fusion, Tabular Transformer, Explainable AI.**

I. Introduction

The integration of IoMT in hospitals, with a primary focus on emergency departments, offers a perspective of accelerating automated triage while improving early disease detection [1]. To train robust machine learning algorithms, large amounts of high-quality, multimodal data are required. However, accessing real-world patient data is heavily restricted by privacy regulations, such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). Such data access is strictly regulated and unauthorized use is punishable by law. Consequently, researchers often lack the data necessary to develop and benchmark innovative multimodal fusion architectures. To address this challenge, synthetic data generation has emerged as the main solution. While real-world data is by nature noisy, incomplete, and legally protected, generating synthetic data at a large scale proves to be highly efficient. It provides an excellent ground for researchers to simulate complex physiological interactions while strictly preserving patient privacy.

This study propose two objectives. First, we propose a simulation engine capable of generating realistic, multimodal synthetic EHRs. This pipeline mimics a two-tier hospital triage system, producing continuous IoMT signals (e.g., SpO2, Heart Rate) fused with discrete laboratory results (e.g., Troponin, D-Dimers) for 24 distinct diseases.

Second, we use this dataset as a benchmark to evaluate two modern Deep Learning paradigms for tabular data:

- A classical Multi-Layer Perceptron (MLP)
- A state-of-the-art Tabular Transformer.

Our contributions are summarized as follows:

- **Multimodal Synthetic Generation:** We engineered a rule-based physiological pipeline to generate a diverse, imbalanced, and sparse cohort of 200,000 synthetic patient records.
- **Deep Learning Benchmarking:** We established predictive baselines on this new dataset by implementing Early Fusion (MLP) and Contextual Fusion (Transformer) architectures.
- **Clinical Validation via XAI:** We utilized SHAP to demonstrate that the synthetic generation rules successfully translate into coherent, medically interpretable decision pathways within the neural networks.

II. Related Work

A. Synthetic Healthcare Data Generation

Overcoming the medical data silo has driven significant research into synthetic generation. Deep Generative models, such as medGAN [2] and Variational Autoencoders (VAEs), are frequently used to synthesize tabular EHRs. While statistically powerful, these purely data-driven models often struggle to maintain strict logical and physiological constraints (e.g., preventing contradictory laboratory values). In contrast, our approach utilizes a rule-based, expert-driven physiological generation pipeline, ensuring absolute clinical coherence across multiple data modalities, making it highly suitable for validating diagnostic algorithms.

B. IoMT and Multimodal Fusion

Recent literature in medical AI emphasizes the necessity of multimodal fusion—synthesizing high-frequency sensor data with low-frequency clinical information [3]. While most IoMT frameworks focus on localized time-series analysis (e.g., detecting arrhythmias) [4], our synthetic dataset explicitly simulates the spatial fusion of these modalities into a unified tabular snapshot, representing the exact moment of emergency triage.

C. Deep Learning on Tabular Data and XAI

Tabular data remains a challenging domain for Deep Learning, historically dominated by tree-based methods. However, the introduction of architectures like the FT-Transformer [5] has shown that projecting tabular features into embedding tokens allows models to capture complex, non-linear interactions via Self-Attention. Furthermore, to ensure trust in these models, SHAP [6] has become the gold standard for Explainable AI (XAI), providing local and global feature importance that we use to validate the physiological integrity of our synthetic data.

D. The Challenge of Imbalanced Clinical Datasets

A pervasive issue in medical Deep Learning is the extreme class imbalance inherent to real-world epidemiological data, where rare, life-threatening conditions are heavily outnumbered by benign cases. Traditional resampling techniques, such as SMOTE [7], often fail in multimodal contexts because they interpolate in the feature space without respecting physiological boundaries, generating clinically impossible synthetic samples. Our approach circumvents this limitation. By leveraging an expert-driven rule-based engine, we directly generate the minority classes (e.g., severe sepsis or hemorrhagic strokes) at scale. Combined with cost-sensitive learning algorithms, such as our weighted Cross-Entropy implementation, this ensures that the neural networks learn accurate, discriminative boundaries for critical conditions without sacrificing the physiological integrity of the tabular representations.

III. Proposed Approach

To address the complexities of multimodal IoMT data fusion and the inherent lack of accessible medical datasets, we designed an end-to-end framework comprising a robust synthetic data generation engine and an advanced Deep Learning evaluation pipeline.

A. Rule-Based Synthetic Data Generation Engine

To simulate a realistic clinical environment, we developed a two-tier generation pipeline capable of mapping multimodal inputs to 24 target pathologies. Tier 1 generates baseline physiological modalities mimicking continuous IoMT devices, including Age, Heart Rate (*FC*), Respiratory Rate (*FR*), and SpO2. Based on specific pathological rules, patients are routed to Tier 2 (Cardiology, Neurology, Pulmonology, and Infectious Diseases), where disease-specific, discrete laboratory biomarkers (e.g., *Troponin*, *D-Dimers*, *Lactates*) are synthesized.

The selection of the 24 target pathologies was not arbitrary; it strictly reflects the most frequent and time-critical admissions in standard Emergency Departments (ED). The dataset spans four major clinical branches: Cardiology (e.g., Acute Coronary Syndromes, Heart Failure), Pulmonology (e.g., Pulmonary Embolism, COPD exacerbations), Neurology (e.g., Ischemic and Hemorrhagic Strokes), and Infectious Diseases (e.g., Severe Sepsis, Bacterial Pneumonia). These conditions were deliberately chosen because their early detection heavily relies on the exact multimodal fusion our architecture proposes: crossing continuous vital anomalies (detectable via IoMT) with specific biomarker elevations (e.g., Troponin for STEMI, D-Dimers for Embolism).

Furthermore, benign conditions with overlapping symptoms, such as Hyperventilation, were included to train the network to handle clinical noise and minimize false-positive triage alerts. To mimic the inherent sparsity of real-world medical data, the resulting arrays are merged using a left-join operation (`combine_first`). If a synthetic patient is admitted for a purely respiratory issue, cardiological markers remain naturally absent. This process yielded a complex, sparse dataset of 200,000 synthetic patient records.

B. Preprocessing and Imbalance Handling

The simulated unperformed medical exams (NaNs) were imputed with zeros, mathematically encoding the clinical decision that the specific test was unnecessary during triage. Continuous variables were standardized using a *StandardScaler* to ensure uniform gradient propagation. To reflect true epidemiological distributions, our generation engine produces heavily imbalanced classes (e.g., rare severe sepsis versus frequent asthma attacks).

We countered this by integrating a balanced Class Weighting strategy directly into the PyTorch Cross-Entropy loss function. For a dataset with C classes, the weighted loss $\mathcal{L}$ for a single patient is defined as:

$$\mathcal{L} = -\sum_{i=1}^{C} w_i y_i \log(\hat{y}_i) \tag{1}$$

Where y_i is the binary indicator of the true class, $\hat{y}_i$ is the predicted probability, and w_i is the inversely proportional weight assigned to class i. This forces the network to penalize false negatives on rare minority classes heavily.

C. Benchmarked Architectures

We benchmarked the synthetic dataset using two distinct data fusion strategies, conceptually illustrated in Figure 1.

- **Baseline MLP (Early Fusion):** A three-layer fully connected network (256, 128, 24 neurons) utilizing Batch Normalization and Dropout [8]. As shown on the left side of Figure 1, it concatenates all synthetic modalities into a single flat input vector before processing, assuming static relationships between features.
- **Tabular Transformer (Contextual Fusion):** Inspired by recent advances in tabular deep learning, this model projects each synthetic feature into a $d_{model} = 64$ embedding space. The tokens interact via Multi-Head Self-Attention ($n\_heads = 4$, $n\_layers = 2$). The attention mechanism allows the network to dynamically weigh the importance of variables based on the context:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

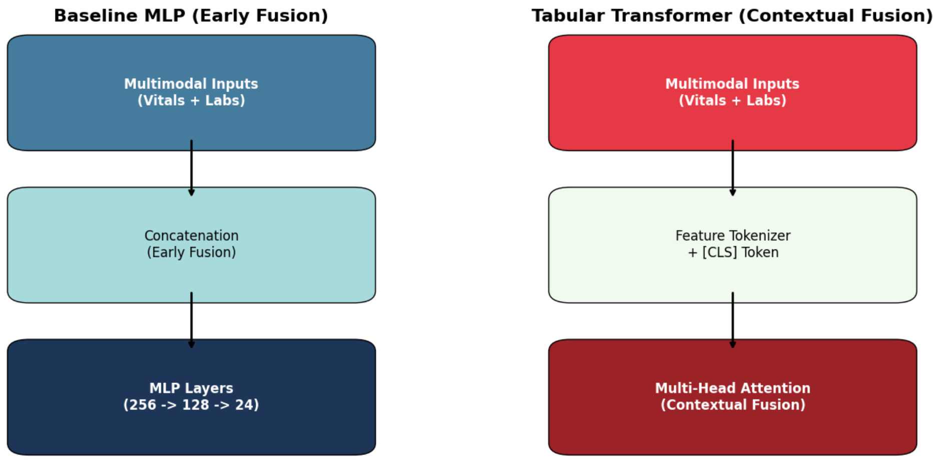

Fig. 1: Block diagram comparing the Early Fusion approach of the MLP with the Contextual Fusion approach of the Tabular Transformer.

This mathematical contextualization (Figure 1, right) mimics a physician analyzing how a specific blood pressure reading changes the interpretation of a heart rate anomaly.

D. *Training Strategy*

Models were optimized using the AdamW optimizer [9] with a decoupled weight decay ($1e-4$) to improve generalization. We implemented a dynamic learning rate scheduler (*ReduceLROnPlateau*) reducing the learning rate by half upon validation stagnation. Finally, an Early Stopping mechanism was employed to halt training and restore the best weights, strictly preventing overfitting on the synthetic clinical noise.

IV. Experimental Results

The evaluation of our multimodal fusion framework was conducted on a 20% hold-out test set derived from the 200,000 synthetic patient records. This section details the validation of the synthetic data, the predictive performance, the computational efficiency, and a deep dive into model interpretability.

A. *Validation of Synthetic Physiological Coherence*

Before evaluating the predictive models, it is crucial to verify that the synthetic generation engine produced realistic clinical relationships rather than random statistical noise. Figure 2 presents the physiological correlation matrix of the generated variables.

The heatmap demonstrates strong positive and negative correlation clusters that align with medical reality. For instance, the correlation block between distinct biomarkers confirms that our rule-based system successfully maintained known physiological dependencies, rendering the dataset a valid, highly structured benchmark for medical Deep Learning.

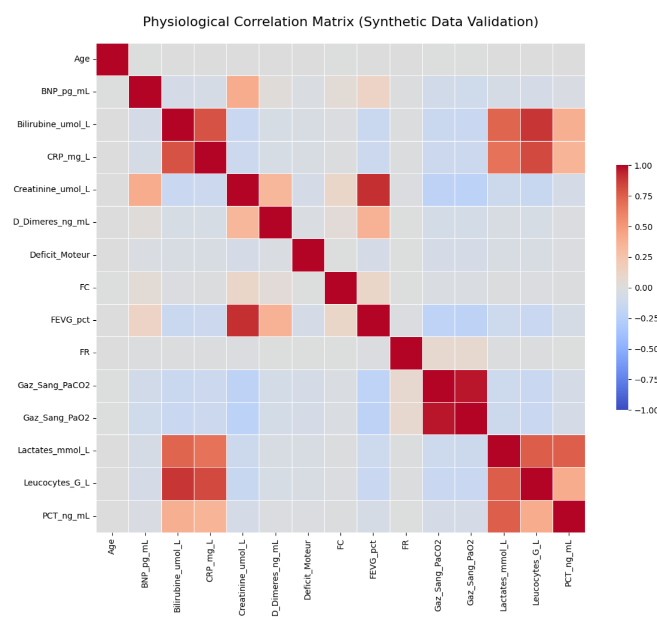

Fig. 2: Physiological Correlation Matrix confirming the realistic inter-variable dependencies within the synthetic IoMT dataset.

B. *Experimental Setup and Hyperparameters*

To ensure reproducibility, the hyperparameter configurations for both architectures are summarized in Table I. Both models were implemented in PyTorch and trained on an optimized hardware environment.

TABLE I: Model Training Hyperparameters

Hyperparameter	MLP Baseline	Tabular Transformer
Optimizer	AdamW	AdamW
Learning Rate	0.002	0.002
Weight Decay	$1e-4$	$1e-4$
Dropout Rate	0.3 / 0.2	0.2
Batch Size	1024	1024
Embedding Dim (d_{model})	N/A	64
Attention Heads	N/A	4

C. Predictive Performance and ROC-AUC Analysis

As illustrated in Figure 3, both models demonstrate high efficiency in classification tasks across the 24 highly imbalanced pathologies.

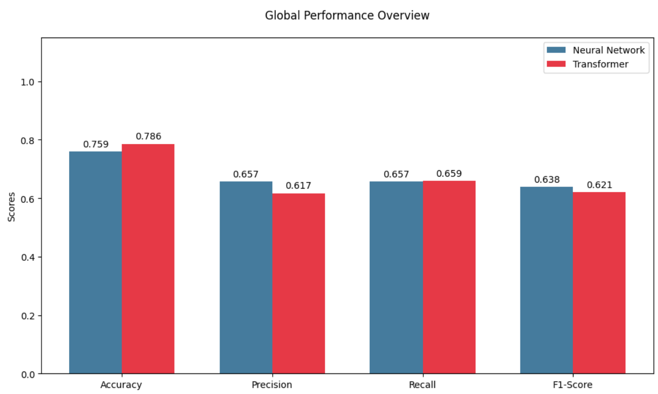

Fig. 3: Global Performance metrics illustrating the trade-offs between Accuracy and Precision across models.

The Tabular Transformer achieved an impressive Global Accuracy of 78.6%, outperforming the Neural Network baseline (75.9%). However, a deeper look at the metrics reveals an interesting dynamic: the Neural Network achieved a slightly better Precision (65.7% vs 61.7%) and a higher Macro F1-Score (63.8% vs 62.1%). This suggests that while the Transformer is highly effective at categorizing the majority classes (boosting overall accuracy), the simpler MLP, aided by class weighting, was slightly more conservative and precise when predicting minority target pathologies.

The discriminative power is further validated by the Receiver Operating Characteristic (ROC) analysis in Figure 4.

Both architectures reached an identical, near-perfect Micro-Average AUC of 0.994. This extremely high score proves that the mathematical signatures generated by our synthetic engine are highly distinct. The models are not randomly guessing but have successfully mapped the multidimensional boundaries of the 24 diseases.

D. Precision-Recall Analysis for Imbalanced Classes

While ROC-AUC is standard, the Precision-Recall (PR) curve provides a more stringent assessment for heavily imbalanced medical datasets.

Figure 5 demonstrates the PR curves for both architectures. The Tabular Transformer (Red) achieves a Micro-Average

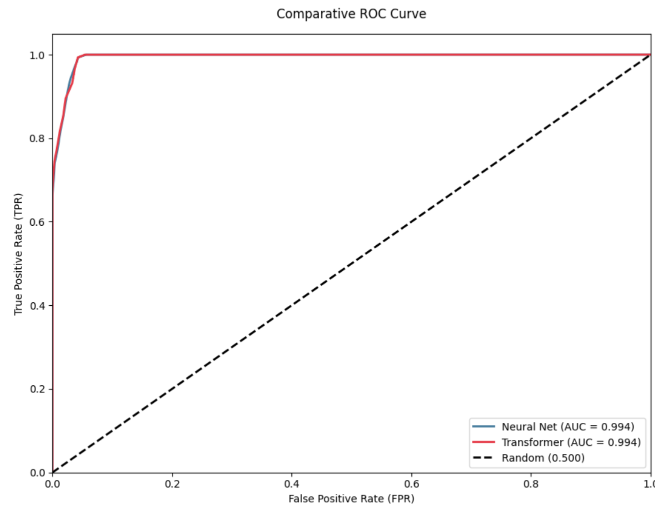

Fig. 4: Comparative ROC Curves demonstrating near-perfect sensitivity and specificity for both models (AUC = 0.994).

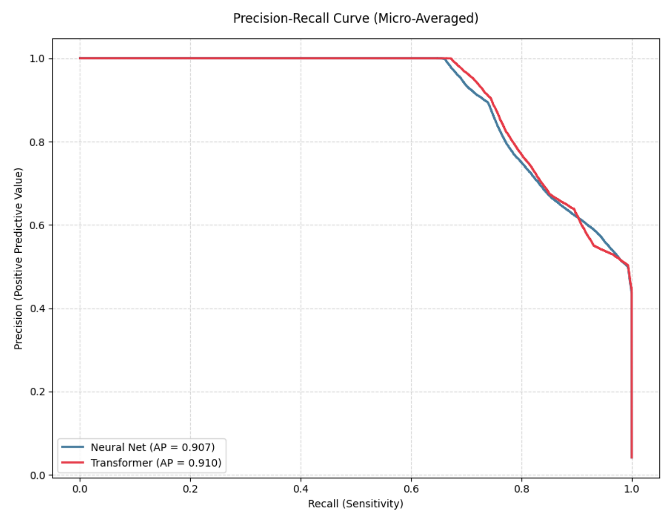

Fig. 5: Precision-Recall (PR) Curve illustrating model robustness on imbalanced clinical classes.

Precision (AP) of 0.910, marginally outperforming the Neural Network (0.907). The curves remain remarkably high across the recall spectrum, confirming that both models can identify a vast majority of critical cases (high sensitivity) without triggering an excessive number of false alarms (high precision).

E. Convergence and Overfitting Control

The stability of the training process was monitored through the Cross-Entropy loss curves shown in Figure 6.

Both architectures demonstrate healthy convergence. The Neural Network (left) converged smoothly over approximately 25 epochs. The Transformer (right) learned aggressively, triggering the Early Stopping mechanism earlier (around 11 epochs). Most importantly, for both models, the Validation Loss (dashed red line) closely tracks or remains below the Training Loss (solid blue line). This confirms that our dropout

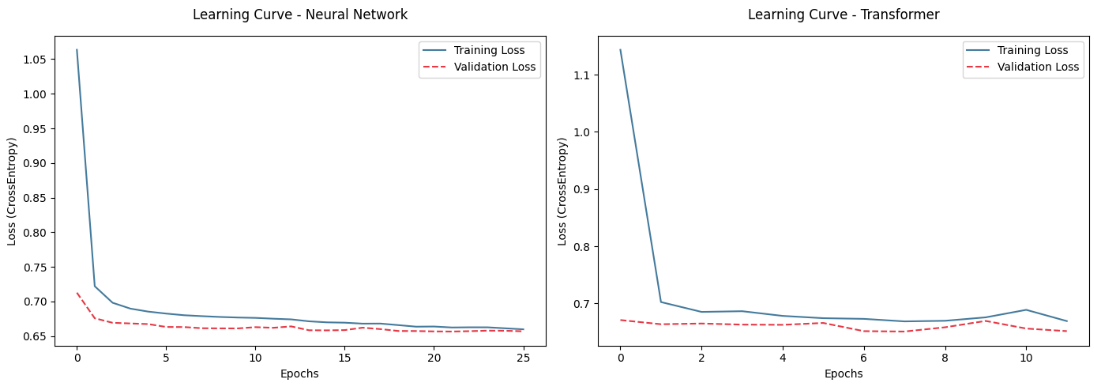

Fig. 6: Learning curves for MLP (left) and Transformer (right) architectures showing stable loss reduction without overfitting.

strategies and Early Stopping successfully prevented the models from memorizing the synthetic dataset.

F. Computational Overhead and Scalability

A major finding of this study relates to the severe trade-off between architectural complexity and computational cost, as highlighted in Figure 7.

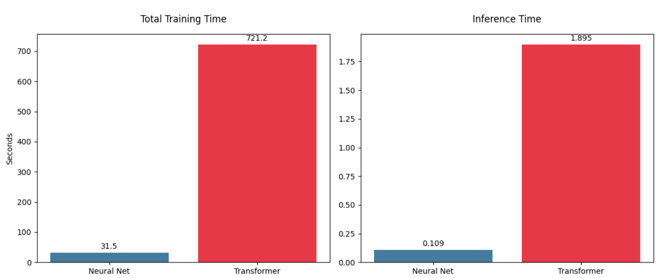

Fig. 7: Total training time (left) and inference latency (right) comparison.

The Transformer required an excessive 721.2 seconds to train, compared to a mere 31.5 seconds for the MLP baseline. This discrepancy is mirrored in inference time: 1.895 seconds for the Transformer versus 0.109 seconds for the MLP. The $O(N^2)$ algorithmic complexity of the Multi-Head Attention mechanism introduces a massive computational penalty. Therefore, while Transformers are powerful for cloud-based hospital servers, the MLP remains the superior architecture for Edge computing and real-time IoMT deployment.

G. Error Analysis and Clinical Overlap

To understand the limitations of the fusion logic, we analyzed the confusion matrices (Figures 8 and 9).

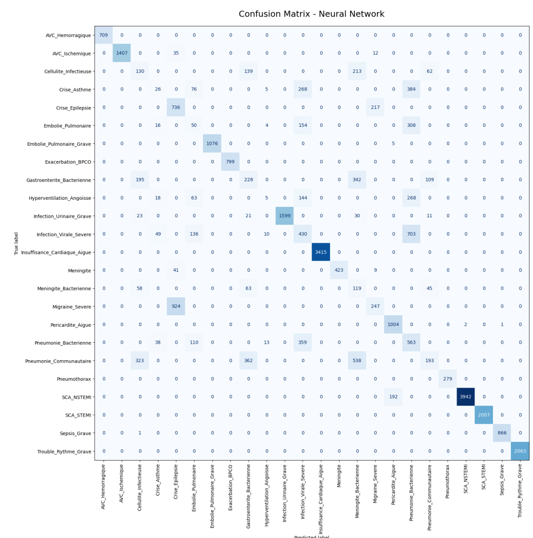

Fig. 8: Confusion Matrix for the Neural Network baseline.

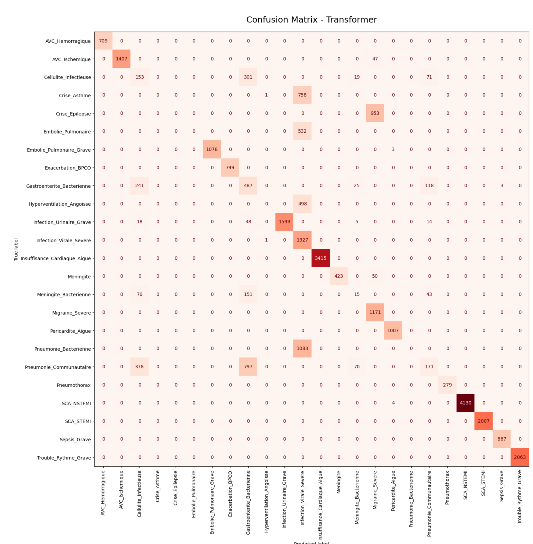

Fig. 9: Confusion Matrix for the Tabular Transformer.

Both matrices show an exceptionally strong diagonal, confirming high accuracy. However, deliberate clinical noise programmed into our synthetic data induced logical misclassifications. For example, both models show slight hesitation between overlapping respiratory and infectious conditions (e.g., distinguishing between different bacterial or viral severities that share similar fever and heart rate profiles). This validates that the synthetic data is non-trivial and correctly mimics the ambiguity found in real-world clinical environments.

V. Conclusion and Future Work

The inability to access comprehensive medical records significantly hinders the development of Deep Learning in healthcare. In this paper, we successfully bridged this gap by developing a robust, rule-based generation engine that produced 200,000 highly realistic, multimodal synthetic patient records simulating an IoMT triage environment. By benchmarking an MLP and a Tabular Transformer on this dataset, we demonstrated that our synthetic data is complex, sparse, and exhibits realistic overlapping clinical phenotypes. Furthermore, the application of SHAP visually validated that the generated data strictly adheres to physiological coherence.

However, a notable limitation of our current framework is the absence of unstructured medical imaging data. In modern clinical settings, radiological imaging—such as Chest X-rays, CT scans, or MRIs—is often the definitive modality for diagnosing conditions like Pulmonary Embolism or Hemorrhagic Stroke.

Therefore, future work will focus on expanding our synthetic generation pipeline to include paired medical images, potentially leveraging advanced Generative AI (e.g., Diffusion Models). Consequently, we aim to upgrade the fusion architecture by integrating Computer Vision models (such as CNNs or Vision Transformers) to extract features from these scans. Concatenating these high-dimensional spatial modalities with our existing tabular IoMT framework will pave the way for a truly comprehensive, multimodal diagnostic assistant.

References

[1] A. Esteva et al., "A guide to deep learning in healthcare," *Nature Medicine*, vol. 25, no. 1, pp. 24-29, 2019.

[2] E. Choi, S. Biswal, B. Malin, J. Duke, W. F. Stewart, and J. Sun, "Generating Multi-label Discrete Patient Records using Generative Adversarial Networks," in *Machine Learning for Healthcare Conference*, PMLR, pp. 286-305, 2017.

[3] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal Machine Learning: A Survey and Taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423-443, 2019.

[4] A. M. Rahmani et al., "Exploiting smart e-Health gateways at the edge of healthcare Internet-of-Things: A fog computing approach," *Future Generation Computer Systems*, vol. 78, pp. 641-658, 2018.

[5] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting Deep Learning Models for Tabular Data," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 18932-18943, 2021.

[6] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[7] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002.

[8] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," *Journal of Machine Learning Research*, vol. 15, no. 1, pp. 1929-1958, 2014.

[9] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," in *International Conference on Learning Representations (ICLR)*, 2019.